Design note: a latent escalation model for conflict and unrest

Question. When unrest rises in a region, does conflict follow *because of* the unrest (escalation), or do both rise together because of a common latent episode (shared episode)? The current joint model separates the two only within four weeks in the same cell (κ). The new model separates them over months and hundreds of kilometres.

0. Input and output

Input (data). Cells $s = 1, \ldots, n$ (0.5°), weeks $t = 1, \ldots, T$.

- Outcomes: $Y^C_{st} = 1$ if at least one conflict event is recorded in cell s, week t, and $Y^U_{st} = 1$ for unrest; 0 otherwise.
- Covariates x_{st} (climate anomalies, population, national indicators, missingness indicators, biome, year), and the country $c(s)$ of each cell.
- Histories, computed from the outcomes themselves: h^C_{st}, $h^U_{st} = \log(1 + \text{number of events in weeks } t-4, \ldots, t-1)$, and their neighbour means $\bar{h}^C_{st}$, $\bar{h}^U_{st}$.
- Grid for the space-time fields: coarse cells $g = g(s)$ (1° or 2°) and time steps $k = k(t)$ of Δ weeks, so all weeks of step k in cell s share the same field value.

Output. (a) Estimates of the parameters listed in Step 4, above all the escalation coefficients $\psi_{U\to C}$, $\psi_{C\to U}$, the shared loading λ and the short-run coefficient κ; (b) maps over time of the three latent fields; (c) predictive probabilities π^C_{st}, π^U_{st} for held-out data; (d) the impulse response of conflict to an unrest episode (Step 5).

1. Data level: a bivariate Bernoulli for each cell-week

Given the two linear predictors η^C_{st} and η^U_{st} (Step 2), the pair (Y^C_{st}, Y^U_{st}) is bivariate Bernoulli with independent components, and pairs are independent across cells and weeks:

$$P(Y^C_{st} = a, Y^U_{st} = b \mid \eta^C_{st}, \eta^U_{st}) = (\pi^C_{st})^a (1-\pi^C_{st})^{1-a} (\pi^U_{st})^b (1-\pi^U_{st})^{1-b}, \qquad \pi^j_{st} = \operatorname{expit}(\eta^j_{st}).$$

All dependence between conflict and unrest therefore comes from the levels below: the histories (lagged, recorded), the shared field (same step) and the escalation terms (lagged, latent). Once the latent fields are integrated out, (Y^C_{st}, Y^U_{st}) are dependent. A same-week odds-ratio parameter at this level is possible, but in the current fits drawing the two outcomes jointly rather than independently changes the joint log score by 0.00002–0.00004, so we do not add it.

2. Linear predictors

$$\eta^C_{st} = \underbrace{x^\top_{st}\beta^C + \delta^C h^C_{st} + \kappa h^U_{st} + \delta^C_{nb}\bar{h}^C_{st} + \kappa_{nb}\bar{h}^U_{st}}_{\text{fixed effects}} + \underbrace{u^C_{c(s)} + e^C_s + w^C_s}_{\text{static random effects}} + \underbrace{v^S_{g,k} + v^C_{g,k}}_{L^C_{g,k}},$$

$$\eta^U_{st} = \underbrace{x^\top_{st}\beta^U + \delta^U h^U_{st} + \kappa^U h^C_{st} + \delta^U_{nb}\bar{h}^U_{st} + \kappa^U_{nb}\bar{h}^C_{st}}_{\text{fixed effects}} + \underbrace{u^U_{c(s)} + e^U_s + w^U_s}_{\text{static random effects}} + \underbrace{\lambda\, v^S_{g,k} + v^U_{g,k}}_{L^U_{g,k}},$$

with $g = g(s)$, $k = k(t)$. We write m^C_{st} and m^U_{st} for the fixed and static random effects together; they are exactly as in the current model. L^C and L^U are the *latent conflict and unrest episodes*: the part of each log-odds that is smooth in space, evolves over time and is not explained by covariates or recorded histories.

3. Latent level, given the hyperparameters

Fixed effects. β^j, δ^j, κ and the neighbour coefficients: one coefficient each, common to all cells and weeks, with vague $\mathcal{N}(0, 10^4)$ priors.

Static random effects (constant over time; one independent set per outcome $j \in \{C, U\}$, as now). These are *random* effects: one value per country or cell, drawn from a distribution whose variance is estimated.

- $u_c^j \sim \mathcal{N}(0, \sigma_{u,j}^2)$, independent over countries: national level.
- $e_s^j \sim \mathcal{N}(0, \sigma_{e,j}^2)$, independent over cells: single-cell level (a town, a border crossing), no spatial correlation.
- $\boldsymbol{w}^j \sim \mathcal{N}(\mathbf{0}, \boldsymbol{Q}_s(\rho_{w,j}, \sigma_{w,j})^{-1})$: a smooth Matérn field with sea barriers ($\boldsymbol{Q}_s$ is the finite-volume SPDE precision of the paper), correlated between nearby cells: regional hotspots that cross borders.

Dynamic terms. Three space-time fields on the coarse grid: v^S (shared), v^C (conflict only), v^U (unrest only). Write $\boldsymbol{D} = (1-\omega)\boldsymbol{I} + \omega\boldsymbol{N}$, where $\boldsymbol{N}$ averages over the neighbours of a coarse cell, so $\boldsymbol{DL}$ is "the episode here and next door". For $k = 2, \ldots, K$:

$$
\begin{aligned}
\boldsymbol{v}_k^S &= \varphi_S \, \boldsymbol{v}_{k-1}^S + \boldsymbol{\varepsilon}_k^S, \\
\boldsymbol{v}_k^C &= \varphi_C \, \boldsymbol{v}_{k-1}^C + \psi_{U\to C} \, \boldsymbol{D} \, \boldsymbol{L}_{k-1}^U + \boldsymbol{\varepsilon}_k^C, \qquad \boldsymbol{L}_{k-1}^U = \lambda \, \boldsymbol{v}_{k-1}^S + \boldsymbol{v}_{k-1}^U, \\
\boldsymbol{v}_k^U &= \varphi_U \, \boldsymbol{v}_{k-1}^U + \psi_{C\to U} \, \boldsymbol{D} \, \boldsymbol{L}_{k-1}^C + \boldsymbol{\varepsilon}_k^U, \qquad \boldsymbol{L}_{k-1}^C = \boldsymbol{v}_{k-1}^S + \boldsymbol{v}_{k-1}^C.
\end{aligned}
$$

The innovations are independent across fields and steps: $\boldsymbol{\varepsilon}_k^\ell \sim \mathcal{N}(\mathbf{0}, \boldsymbol{Q}_s(\rho_\ell, \sigma_\ell)^{-1})$, $\ell \in \{S, C, U\}$, each a Matérn field with its own range ρ_ℓ and innovation SD σ_ℓ. In matrix form, $\boldsymbol{z}_k = (\boldsymbol{v}_k^S, \boldsymbol{v}_k^C, \boldsymbol{v}_k^U)$ is a vector autoregression $\boldsymbol{z}_k = \boldsymbol{\Phi}\, \boldsymbol{z}_{k-1} + \boldsymbol{\varepsilon}_k$ with

$$
\boldsymbol{\Phi} = \begin{pmatrix} \varphi_S \boldsymbol{I} & \mathbf{0} & \mathbf{0} \\ \psi_{U\to C} \lambda \, \boldsymbol{D} & \varphi_C \boldsymbol{I} & \psi_{U\to C} \boldsymbol{D} \\ \psi_{C\to U} \boldsymbol{D} & \psi_{C\to U} \boldsymbol{D} & \varphi_U \boldsymbol{I} \end{pmatrix}.
$$

The first step $\boldsymbol{z}_1$ has the stationary variance of the uncoupled fields, $\boldsymbol{v}_1^\ell \sim \mathcal{N}(\mathbf{0}, \boldsymbol{Q}_s(\rho_\ell, \sigma_\ell)^{-1} / (1 - \varphi_\ell^2))$. *Stationarity*: the spectral radius of $\boldsymbol{\Phi}$ must be below one; we check it at the estimate. *Precision*: stacking $\boldsymbol{z} = (\boldsymbol{z}_1, \ldots, \boldsymbol{z}_K)$, $\boldsymbol{Q}_z = \boldsymbol{B}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{B}$, where $\boldsymbol{B}$ has identity blocks on the diagonal and $-\boldsymbol{\Phi}$ below it, and $\boldsymbol{\Sigma}^{-1}$ is block diagonal in the $\boldsymbol{Q}_s$. $\boldsymbol{Q}_z$ is sparse, and since $\det \boldsymbol{B} = 1$, $\log\det \boldsymbol{Q}_z$ does not depend on λ, ψ or ω, as with the loading in the current model.

4. Hyperparameters and priors

static, per outcome (8)	$\sigma_{u,j}, \sigma_{e,j}, \rho_{w,j}, \sigma_{w,j}$	PC priors, as now
dynamic fields (9)	$\rho_\ell, \sigma_\ell, \varphi_\ell$, $\ell \in \{S, C, U\}$	PC priors; $\operatorname{atanh} \varphi_\ell \sim \mathcal{N}(0, 1.5^2)$
shared loading (1)	λ	$\mathcal{N}(0, 2^2)$
escalation (2)	$\psi_{U\to C}$, $\psi_{C\to U}$	$\mathcal{N}(0, 0.5^2)$
spatial reach (1)	$\omega \in (0,1)$	$\operatorname{logit} \omega \sim \mathcal{N}(0,1)$, or fixed at 0.5

21 hyperparameters, against 15 in the current extended model (8 static, shared $\rho, \sigma, \varphi, \lambda$, unrest-only ρ, σ, φ).

5. What each parameter answers, and the special cases

- λ: how strongly a shared episode moves unrest relative to conflict, in the same step and place. φ_S: how long shared episodes last.
- $\psi_{U\to C}$: latent escalation. A one-unit rise in the unrest episode in step k, here and next door, raises the conflict log-odds by $\psi_{U\to C}$ in step $k+1$, and the effect then decays at rate φ_C. $\psi_{C\to U}$: the reverse direction.
- κ: short-run escalation from *recorded* unrest in the last four weeks, same cell, as now.
- *Impulse response*: the change in conflict log-odds $h = 0, 1, 2, \ldots$ steps after a one-SD innovation in v^S (shared part: immediate in both outcomes) or in v^U (escalation part: zero at $h = 0$, then through $\psi_{U\to C}$). Computed from $\boldsymbol{\Phi}^h$.

Special cases. *Current extended model*: $\psi_{U\to C} = \psi_{C\to U} = 0$ and $v^C \equiv 0$. *Current joint model*: in addition $v^U \equiv 0$. *Univariate space-time model*: only $m^C + v^S$, no unrest data.

6. Inference

Empirical Bayes with a Laplace approximation, as now. Given the hyperparameters $\boldsymbol{\theta}$, all latent quantities $\boldsymbol{\xi} = (\boldsymbol{\beta}, \boldsymbol{\delta}, \kappa, \boldsymbol{u}, \boldsymbol{e}, \boldsymbol{w}, \boldsymbol{z})$ are jointly Gaussian with sparse precision; the Laplace approximation of $p(\boldsymbol{y} \mid \boldsymbol{\theta})$ is maximised over $\boldsymbol{\theta}$, and the latent fields are summarised by their conditional posterior at $\hat{\boldsymbol{\theta}}$. The design matrix does not depend on $\boldsymbol{\theta}$ (λ and ψ enter only $\boldsymbol{Q}_z$). Case-control subsampling of non-events ($r = 0.25$) with offset $-\log r$, as now.

Identification. Shared episodes move both outcomes in the *same* step; escalation moves conflict *one step after* unrest. They are separated by timing, so Δ must be short: $\Delta = 4$ weeks on 1° cells for West Africa (91 steps) and $\Delta = 8$ on 2° cells for Africa and the Middle East (46 steps), about $1.3\text{–}1.8 \times 10^5$ latent values each.

Simulation check before any real-data fit. 20 replicates for each truth: (i) shared episodes only; (ii) escalation only ($\psi_{U\to C} = 0.3$); (iii) both; (iv) short-run κ only. Criteria: bias and coverage of $\psi_{U\to C}$, λ, κ; no spurious ψ under (i) and (iv). If ψ and λ are not separated, we stop and report regional λ and κ instead.